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NATIONAL CENTRE FOR TECHNOLOGY MANAGEMENT

INTEGRATED FRAMEWORK FOR R&D COMMERCIALISATION AT PUBLIC RESEARCH INSTITUTIONS

Presented by

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at

NACETEM RESEARCHERS' SEMINAR
SERIES. 20TH MARCH 2025

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- Introduction
- Methodology
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Introduction

- ❖ The *foundation of most advanced economies* are the R&D outputs of their research institutions (RIs). Africa's hope is currently hinged on its public research institutions (PRIs) more than is being acknowledged.
- ❖ PRIs were established to provide R&D outputs relevant to socio-economic development of the country. Hence, the society ultimately values PRIs based on *how many R&D outputs they commercialised* and make available to industries & end-users.
- ❖ A major indicator of performance for PRIs is R&D outputs i.e. developmental outputs, academic publications, human capacity development, quantity and quality of innovations. Majority of organisations, especially PRIs in Africa, struggles with being innovative (Appiah *et al.*, 2012). Nigeria ranks 113th on the GII (WIPO, 2024).

Top 20 Countries & Nigeria	Research Resources		Research Facilities	
	R&D Expenditure (% of GDP) in 2022	R&D Personnel (per million people from 2000-2022)	Universities 2022	Research Institutes 2022
United States	3.59%	4452	3,295	741
China	2.55%	1687	2,313	96
Japan	3.41%	5638	986	71
Germany	3.13%	5536	455	305
South Korea	5.21%	9082	378	25
India	0.65%	260	4,029	506
France	2.18%	5175	569	210
Russia	1.00%	2698	1352	152
United Kingdom	2.77%	4491	287	142
Brazil	1.15%	888	1,539	60
Canada	1.55%	5076	352	117
Australia	1.68%	4894	202	60
Italy	1.45%	2678	236	57
Taiwan	3.96%	9160	158	12
Spain	1.44%	3252	250	58
Netherlands	2.30%	6074	138	67
Sweden	3.41%	8131	47	39
Turkey	1.32%	2000	187	17
Switzerland	3.16%	6023	100	66
Singapore	2.16%	7225	44	9
Nigeria	0.13%	23	260	70



Introduction – contd.

- ❖ R&D: comprise *creative work* undertaken on a systematic basis in order to increase the stock of knowledge (OECD).
- ❖ Innovation: is the search for and the discovery, development, improvement, adoption, and commercialisation of new processes, new products and new organizational structures and procedures introduced by a person or organisation, where such is *perceived to be new*.
- ❖ R&D Commercialisation: is *a process* of transforming research findings into marketable products or services ultimately bringing them to the market for financial gain.





Prior Works & Gaps

- ❖ Several authors have written about R&D commercialisation. Some of the works consulted for this study include Oyelaran-Oyeyinka *et al.* (1996), Erik Arnold and Katalin Balazs (1998), Okebukola (2002), Appiah *et al.* (2012), Siyanbola *et al.* (2008; 2012), Etatuve (2018), etc.
- ❖ There are still gaps in the literature on PRIs' R&D commercialisation. Some of the gaps noticed include not addressing fundamental weaknesses in mandates of PRIs, overdependence on external agents, innovation process chasm between PRIs & the market, etc.
- ❖ This study is based on past literatures; it was partly presented at GLOBELICS 2018.





Problem Statements

- ❖ PRIs are busy working and creating S&T advancements of various kinds. Yet, all these efforts mostly end up in their round-table discussions, conferences, published papers, or worse still in a file cabinet, shelf or dark corners of their laboratories.
- ❖ Commercialisation of R&D outcomes by PRIs is abysmally low (Appiah *et al.*, 2012) due to several factors. A prominent drawback for PRIs in Nigeria is the uncanny perception that they were created not for profit purpose but merely for public good.
- ❖ Most PRIs are not well positioned to commercialise their R&D due to lack of knowledge on basic commercialisation principles, existing culture, incentives and delivery systems (Kalaitzandonakes, 1997); lack of interest in rights protection and consequently commercial exploitation of their results (Appiah *et al.*, 2012).



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Objectives of the Study

The main objectives of this study are to review past R&D commercialisation experiences and models as well as to design a new integrated framework.





Justifications of the Study

- ❖ In developing countries such as Nigeria where major share of the R&D budget for PRIs comes from the government, a fundamental concern is how to provide policies, incentives and enabling conditions for the commercialisation of research outputs for revenue generation, economic growth and industrial development.
- ❖ Majority of the country's efforts on R&D are carried out by tertiary education institutions and PRIs, yet the nature of enabling environment available to PRIs for effective commercialisation of R&D results is not ideal (Siyanbola *et.al*, 2012). This necessitates the development of an integrated model for their R&D commercialisation.



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Methodology of the Study

**TPP of R&D
Commercialisation**

**Commercialisation
Review Models**

**R&D
Commercialisation
Study**

**Integrated
Commercialisation
Framework**

Figure1: Framework for the Study on R&D Commercialisation Model

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Sampling Details:

- ❖ 10 PRIs purposively sampled in 4 states (Lagos, Ogun, Oyo & Edo) in 3 Federal Ministries (FMARD, FMH, FMIST).
- ❖ A questionnaire survey instrument was administered to top management of the PRIs in 2018.
- ❖ Data gathered were analysed using descriptive & inferential statistics. The relationships between the theoretical constructs and the main dependent indicator were defined statistically.
- ❖ An integrated model for commercialisation of R&D by PRIs was developed from the study. It comprise 6 main indicators.





Table2: Integrated Model Design

Indicators	Variables
Institutional Competence	Personnel competence: *R&D staff education profile
	External competence: *Collaborations *Technology transfer *Technology scanning
	Technology competence: *Tool adoption period *R&D productivity/outputs
	Internal processes: *Identifying commercialisable R&D *Supporting commercialisable R&D *Motivating commercialisable R&D
R&D Outputs	*Programs *Products *Publications *Services
Commercialisation Channels	Information/Intellectual platforms: *Databases *Conferences *Media
	Institutional platforms: *Patents *Licenses *Grants
	Commercial platforms: *Spin-off *JVs *Product offerings
Commercialisation Capability	Strategic capability: *R&D staff *R&D commercialisation mandate *R&D commercialisation budget *R&D commercialisation strategy *Commercialisation agent *Commercialisation training
	Contextual capability: *Competitive environment *Commercial environment
	Linkage capability: *Collaborations on commercialisation *Market scanning *R&D commercialisation support organisations
Commercialisation Agents	*Internal vs. External *Foreign vs. Local *Public vs. Private
Market	*Fast Evolving *Several Players *Open *High Barrier *Several Segments



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RESULTS & DISCUSSIONS



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Theory, Policy & Practice (TPP) of Commercialisation

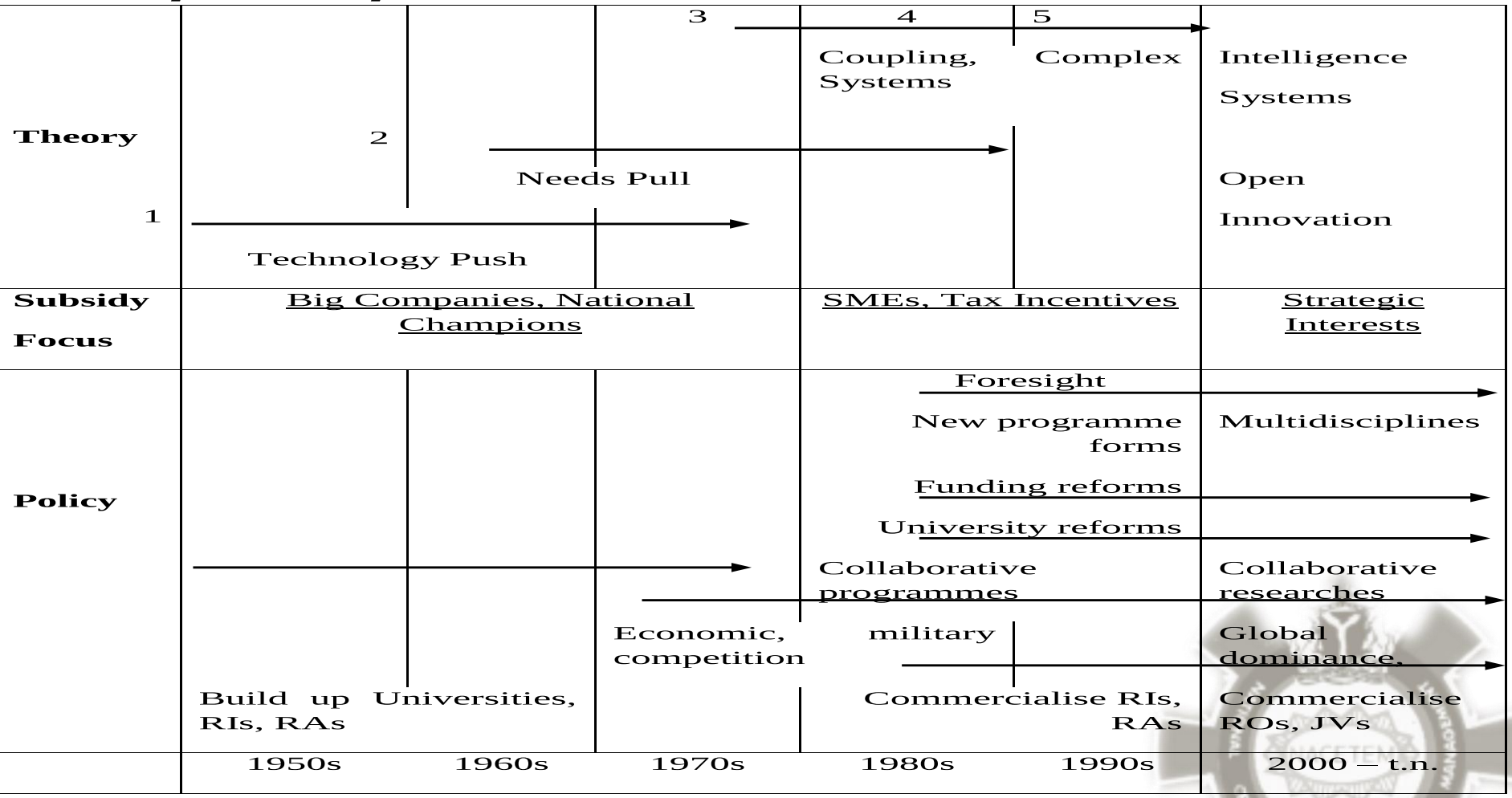


Figure 2: Innovation in Theory and Policy (Source: Erik Arnold and Katalin Balazs, 1998. Framework modified by Babalola Olutunde, 2025).

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TPP of R&D Commercialisation

- ❖ R&D Policy in Nigeria comprise government support for R&D through policies such as R&D funding at MDAs & HEIs; creation of PRIs; intellectual property rights (IPR) administration; and review of the country's S&T policy in 2011 (*innovation* was introduced into the lexicon of the Nigerian S&T policy, now Nigerian STI policy).
- ❖ R&D Practice in Nigeria comprise about 260 public and private Universities and about 70 PRIs in several ministries. The FMIST has about 17 agencies with 14 having mandates for basic/pure research and 3 (NBTI, NACETEM, NOTAP) with mandates for managing, promoting, building & servicing STI functions as well as applied research.



Commercialisation Model Review

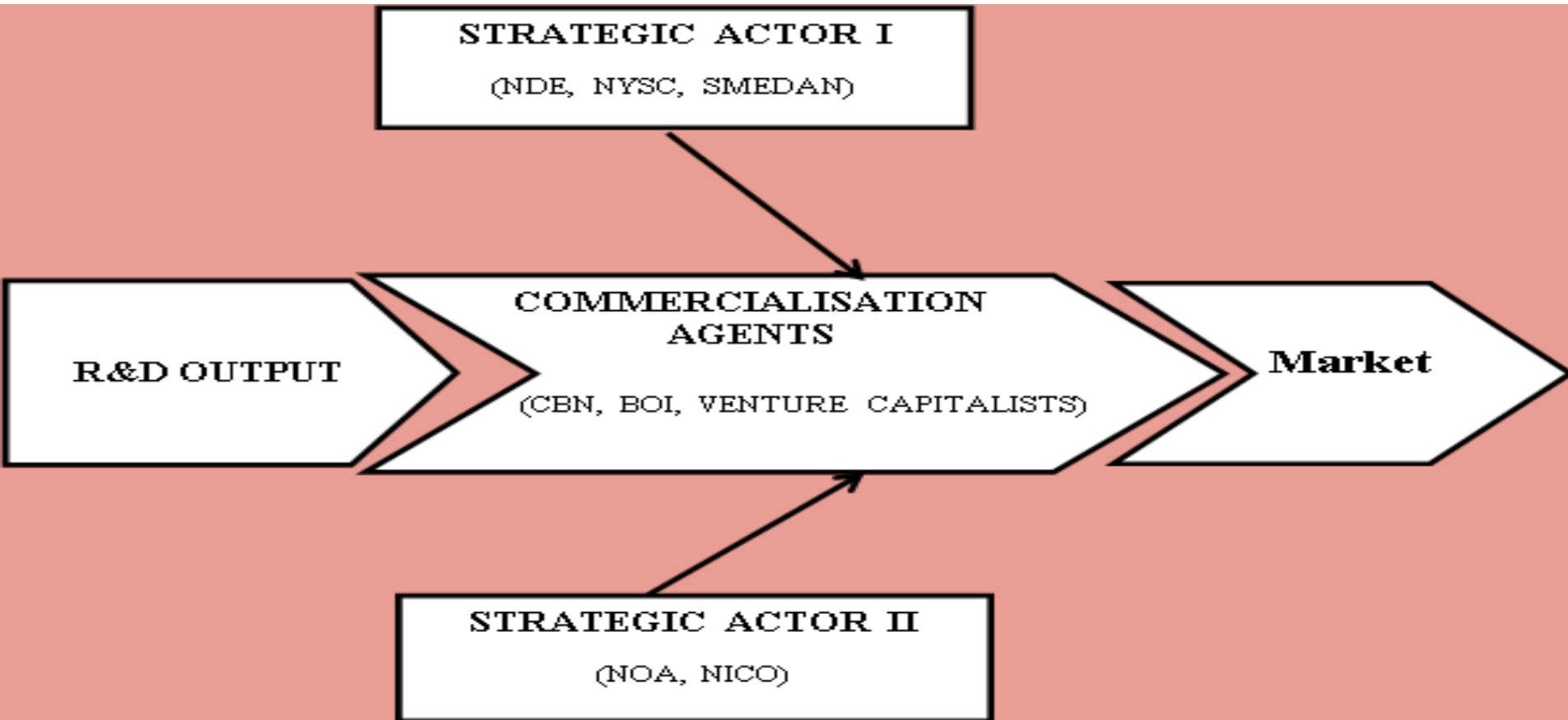


Figure 3: R&D commercialisation in Nigeria - idealised situation model.
(Source: Siyanbola *et al.*, 2012)



R&D Commercialisation Study

- ❖ R&D commercialisation resources (R&D staff, commercialisation budget)
- ❖ Table 3: R&D Commercialization Mandate & Strategies of Public Research Institutes

	Frequency	%
Mandate specifying R&D commercialization:		
Yes	3	30
No	6	60
Nature of Commercialization Strategies:		
Disclosure Platforms (i.e. databases)	7	70
Institutional Platforms (i.e. patenting, licensing)	6	60
Commercial Platforms (i.e. joint ventures, spin off)	4	40



Table 4: R&D Commercialisation Agents Utilized by Public Research Institutes

Details	Frequency	%
Private Venture Capitalists	1	10
Public Venture Capitalists	1	10
Internal Staff (marketing personnel)	5	50
External orientation agencies (NOA, NOTAP, etc.)	2	20
Local bridging institutions (MAN, SMEDAN, etc.)	3	30
Foreign bridging institutions	0	0

**Table 5: Trainings on R&D Commercialisation**

	Frequency	%
Staff Received Trainings:		
Yes	5	50
No	5	50
Nature of Commercialization Training:		
Formal Internal Training	1	10
Formal External Training	4	40
Informal Internal Training	3	30
Informal External Training	1	10

**Table 6: Collaborations for R&D Commercialisation by Public Research**

	Frequency	%
Commercialisation Collaborative Activities:		
Yes	7	70
No	3	30
Nature of Collaborative Activities:		
Information	3	30
Funds	2	20
Technology Sharing	3	30
Training	6	60
Marketing	0	0



Table 7: Internal Mechanisms for R&D Commercialisation

	Frequency	%
Market Scanning:		
Yes	7	70
No	3	30
Support Mechanism:		
Identifying Commercialisable R&D Outputs	7	70
Supporting R&D Outputs Commercialization	9	90
Incentivizing R&D Outputs Commercialization	4	40



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Proposed Integrated Framework

- ❖ Institutional Competence for R&D
- ❖ R&D Outputs
- ❖ R&D Commercialisation Channels
- ❖ R&D Commercialisation Capability
- ❖ R&D Commercialisation Agents
- ❖ Market (Definition)

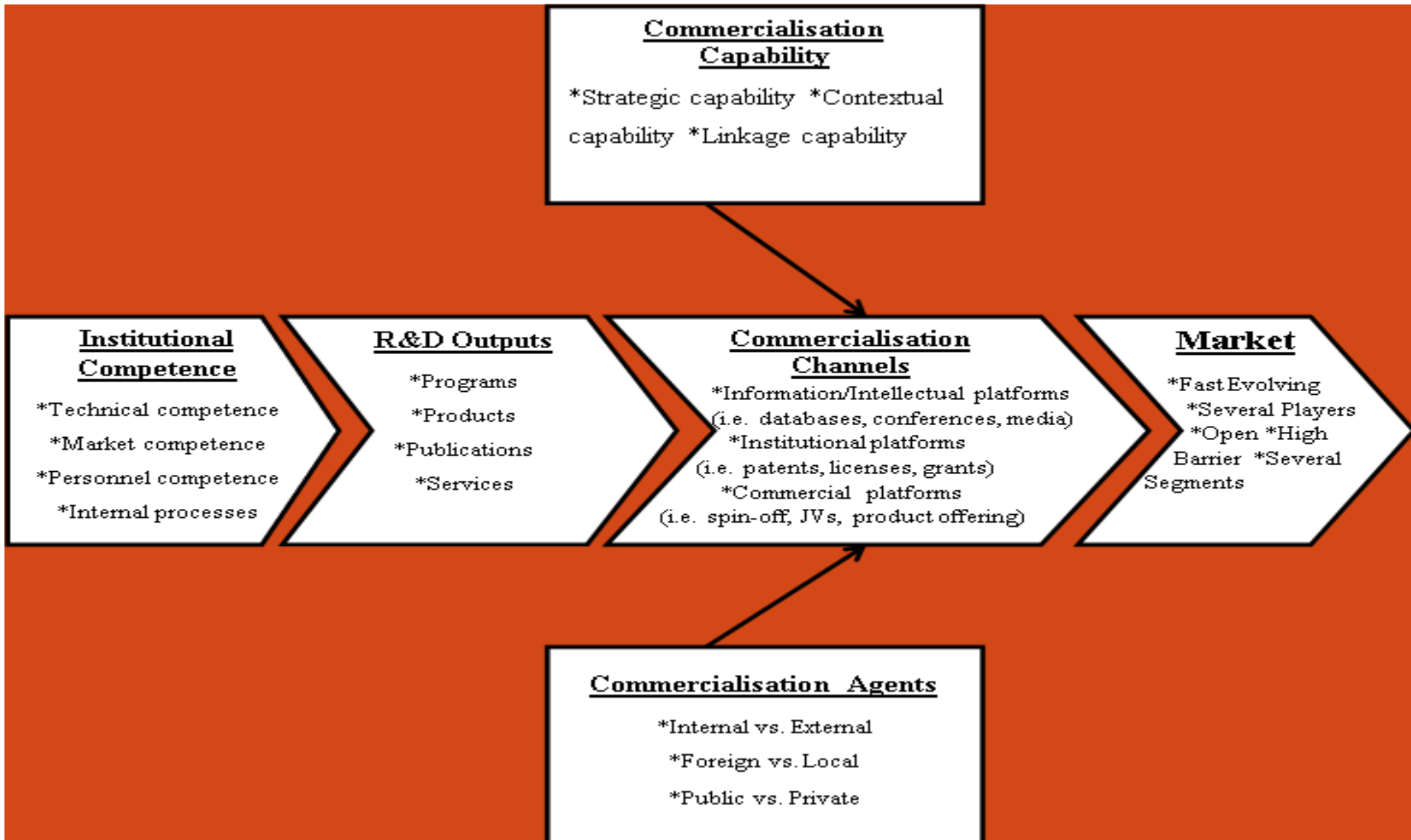




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Integrated Commercialisation Framework





Conclusions

This study examined past theories, models, contexts, and practices of PRIs in regards to R&D commercialisation leading to extraction of critical information necessary for the designing a more contemporary model.

The proposed integrated framework bridged innovation process chasm between PRIs and the market thereby restoring strategic initiative back to the PRIs.





Recommendations

The integrated framework for R&D commercialisation should be put into practice by PRIs adopting it. NACETEM should leverage the framework to build a training program for other PRIs on R&D commercialisation.

NACETEM should start compiling and maintaining R&D and innovation indicators database for Nigeria which, potentially, can help the country improve its ranking on the global innovation index (GII).





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THANK YOU FOR YOUR ATTENTION



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Acknowledgements

This is to appreciate the contributions of the following to the conceptualization and execution of this study.

- ❖ Dr. Etatuve S.
- ❖ Mrs. Dickson F.O.
- ❖ Mr. Amiolemen S.O.





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